Fractional Burgers' Equation

The L1/Crank-Nicolson Predictor-Corrector Scheme

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The Problem: Time-Fractional Burgers' Equation

Our goal is to find a numerical solution for $u(x,t)$ in the following equation:

The Governing Equation

$$\underbrace{u_t^{\alpha}}_{\substack{\text{Caputo Fractional} \\ \text{Derivative (Memory)}}} + auu_x - cu_{xx} = 0, \quad (0 < \alpha \leq 1)$$

- This equation models diffusion and shock waves, but with **memory effects**.
- We need a numerical method that can handle the **fractional derivative** and the **nonlinearity**.

Our Approach: A 3-Part Strategy

We will combine three powerful techniques to build a stable and accurate scheme:

1. **The L1 Formula:** A robust finite difference approximation for the Caputo fractional derivative (u_t^α). This handles the "memory".
2. **Crank-Nicolson (C-N) Scheme:** An implicit, second-order accurate method for the spatial derivatives (u_{xx}). This provides excellent stability.
3. **Predictor-Corrector Method:** A technique to linearize the nonlinear term (auu_x), allowing us to solve the system.

Result

This combination creates a scheme that is unconditionally stable and achieves high accuracy ($O(\Delta t^{2-\alpha}, \Delta x^2)$).

1. Stability Analysis (The "Safety" Claim)

Definition: Stability determines whether numerical errors dampen out over time or grow uncontrollably ("blow up").

Conditional Stability (Standard Methods)

- **Restriction:** You must keep the time step Δt very small relative to Δx.
- **Risk:** If Δt is too large, the solution oscillates wildly or goes to infinity.

Unconditional Stability (Our Result)

- **The "Gold Standard":** The solution remains bounded regardless of the choices for Δt and Δx.
- **Benefit:** We can use larger time steps to save computational time without worrying about the simulation crashing.

2. Accuracy & Convergence: $O(\Delta t^{2-\alpha}, \Delta x^2)$

This term describes how fast the error shrinks as we refine the grid.

- **Space: $O(\Delta x^2)$ (Second-Order)**
  - Halving the grid spacing ($\Delta x \to \Delta x/2$) drops the error by a factor of 4 (2^2).
  - Typical for Central Difference schemes.
- **Time: $O(\Delta t^{2-\alpha})$ (Fractional Order)**
  - Unique to Fractional Calculus. The convergence depends on α.
  - If $\alpha = 1$ (Standard Heat Eq): Order is $2 - 1 = 1$.
  - **If $\alpha = 0.5$:** Order is $2 - 0.5 = 1.5$.

Validation from Future Results

From our future calculations, the numerical order $p \approx \mathbf{1.49}$. This will match the theoretical prediction exactly:

$$\text{Theoretical Order} = 2 - \alpha = 1.5$$

This confirms our method is functioning correctly.

Step 1: The Grid & L1 Formula

First, we discretize the domain:

- $x_j = j\Delta x$ and $t_n = n\Delta t$.
- Our goal is to find $u_j^n \approx u(x_j, t_n)$.

L1 Formula for the Fractional Derivative

We approximate the Caputo derivative at time t_{n+1} using the L1 formula:

$$D_t^\alpha u_j^{n+1} \approx K \sum_{k=0}^{n} b_k (u_j^{n+1-k} - u_j^{n-k})$$

Constants:

- $K = \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)}$
- $b_k = (k+1)^{1-\alpha} - k^{1-\alpha}$ (These are the "memory" weights)

Step 2: Crank-Nicolson (C-N) Scheme

The C-N method "averages" the spatial derivatives between the current step (t_n) and the next step (t_{n+1}).

Viscous Term (cu_{xx})

$$cu_{xx} \approx \frac{c}{2}\left[\delta_x^2 u_j^{n+1} + \delta_x^2 u_j^n\right]$$

where $\delta_x^2 u_j = \frac{u_{j+1} - 2u_j + u_{j-1}}{(\Delta x)^2}$

Nonlinear Term (auu_x)

$$auu_x \approx \frac{a}{2}\left[(uu_x)_j^{n+1} + (uu_x)_j^n\right]$$

The Problem

The term $(uu_x)_j^{n+1}$ is **implicit and nonlinear**. We can't solve this directly.

Step 3: Predictor-Corrector Linearization

We linearize the nonlinear term $(uu_x)_j^{n+1}$ in two steps:

1. **Predict:** We "predict" the value of u in the term. The simplest predictor is the value from the previous time step:

$$u_j^{n+1} \approx u_j^n$$

2. **Correct:** We use this prediction to linearize the full term:

$$(uu_x)_j^{n+1} \approx u_j^{n+1}(\delta_x u_j^{n+1}) \approx u_j^n \left(\frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} \right)$$

The Final Linearized Term

$$auu_x \approx \frac{a}{2} \left[u_j^n \left(\frac{u_{j+1}^{n+1} - u_{j-1}^{n+1}}{2\Delta x} \right) + u_j^n \left(\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \right) \right]$$

Now, the equation is fully linear with respect to the unknown u^{n+1} terms!

The Final Iterative System

Substituting all approximations into $u_t^\alpha = -auu_x + cu_{xx}$ and grouping terms gives a **tridiagonal system**:

The Tridiagonal System

$$A_j u_{j-1}^{n+1} + B_j u_j^{n+1} + C_j u_{j+1}^{n+1} = R_j^n$$

LHS Coefficients (for u^{n+1})

These are the "unknown" terms we solve for.

- $A_j = -\frac{au_j^n}{4\Delta x} - \frac{c}{2(\Delta x)^2}$
- $B_j = \underbrace{\frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)}}_{\text{from L1}} + \underbrace{\frac{c}{(\Delta x)^2}}_{\text{from C-N}}$
- $C_j = \frac{au_j^n}{4\Delta x} - \frac{c}{2(\Delta x)^2}$

The Final Iterative System (RHS)

The RHS R_j^n is calculated entirely from **known values** at previous time steps.

RHS Vector ($R_j^n = H_j^n + S_j^n$)

1. The Fractional "History" Term (H_j^n):

$$H_j^n = K \left[u_j^n - \sum_{k=1}^{n} b_k (u_j^{n+1-k} - u_j^{n-k}) \right]$$

(Note: $K = \frac{(\Delta t)^{-\alpha}}{\Gamma(2-\alpha)}$)

2. The C-N "Source" Term (S_j^n):

$$S_j^n = -\frac{a}{2} u_j^n \left(\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} \right) + \frac{c}{2} \left(\frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{(\Delta x)^2} \right)$$

Key Concept: The "Memory" Effect

How does the next step depend on the past?

- **Integer-Order PDE ($\alpha = 1$):** The next step u^{n+1} depends **only** on the current step u^n.
- **Fractional-Order PDE ($\alpha < 1$):** The next step u^{n+1} depends on **all previous steps** $(u^n, u^{n-1}, \ldots, u^0)$.

The History Term H_j^n

This dependency is captured in the history sum:

$$\sum_{k=1}^{n} b_k (u_j^{n+1-k} - u_j^{n-k})$$

This sum grows longer at every time step, making the computation more expensive as the simulation runs.

Worked Example: Setup

Let's solve for 2 steps with a very coarse grid.

Parameters

- $\alpha = 0.5$
- $a = 1.0$
- $c = 1.0$

Grid

- $\Delta x = 0.5$ (3 points: x_0, x_1, x_2)
- $\Delta t = 0.5$ (We'll find u^1, u^2)

Conditions

$u(x, 0) = x$ (Initial)

$u_0^0 = 0, u_1^0 = 0.5, u_2^0 = 1.0$

$u(0, t) = 0, \quad u(1, t) = 1$ (Boundary)

$u_0^n = 0.0, \quad u_2^n = 1.0$ for all n

Goal

We only need to solve for the single internal point: u_1^n. The system $A_1 u_0^{n+1} + B_1 u_1^{n+1} + C_1 u_2^{n+1} = R_1^n$ is just one equation.

Example: Step 1 $\rightarrow$ Solve for u_1^1 (at $t = 0.5$)

We set $n = 0$. We use u^0 values to find u^1.

1. **Calculate LHS Coeffs (using $u_1^0 = 0.5$):**
   - $A_1 = -(0.5 \times 0.5) - 2.0 = -2.25$
   - $B_1 = 1.596 + 4.0 = 5.596$
   - $C_1 = (0.5 \times 0.5) - 2.0 = -1.75$
2. **Calculate RHS R_1^0 (using u^0):**
   - $H_1^0 = K \times u_1^0 = 1.596 \times 0.5 = 0.798$ (History sum is empty)
   - $S_1^0 = -0.25 \times (1.0) + 0.5 \times (0.0) = -0.25$
   - $R_1^0 = 0.798 - 0.25 = 0.548$
3. **Solve for u_1^1:**
   - $B_1 u_1^1 = R_1^0 - A_1 u_0^1 - C_1 u_2^1$
   - $5.596\, u_1^1 = 0.548 - (-2.25)(0.0) - (-1.75)(1.0)$
   - $5.596\, u_1^1 = 0.548 + 1.75 = 2.298$
   - $u_1^1 \approx 0.4106$

Example: Step 2 $\rightarrow$ Solve for u_1^2 (at $t = 1.0$)

We set $n = 1$. We use u^1 and u^0 values to find u^2.

1. **Calculate LHS Coeffs (using $u_1^1 = 0.4106$):**
   - $A_1 = -(0.5 \times 0.4106) - 2.0 = -2.2053$
   - $B_1 = 5.596$ (Constant)
   - $C_1 = (0.5 \times 0.4106) - 2.0 = -1.7947$
2. **Calculate RHS R_1^1 (using u^1, u^0):**
   - $b_1 = 2^{0.5} - 1^{0.5} \approx 0.414$
   - $H_1^1 = K[u_1^1 - b_1(u_1^1 - u_1^0)] \approx 0.7144$
   - $S_1^1 \approx 0.1523$
   - $R_1^1 = 0.7144 + 0.1523 = 0.8667$
3. **Solve for u_1^2:**
   - $B_1 u_1^2 = R_1^1 - A_1 u_0^2 - C_1 u_2^2$
   - $5.596\, u_1^2 = 0.8667 - (-2.2053)(0.0) - (-1.7947)(1.0)$
   - $5.596\, u_1^2 = 0.8667 + 1.7947 = 2.6614$
   - $u_1^2 \approx 0.4756$

Example: Summary of Results

Here are the final computed values for our single internal point u_1 ($x = 0.5$).

Table: Numerical solution at $x = 0.5$

Time Step	$j = 0$ $(x = 0.0)$	$\mathbf{j = 1}$ $\mathbf{(x = 0.5)}$	$j = 2$ $(x = 1.0)$
u^0 $(t = 0.0)$	0.0	**0.5000**	1.0
u^1 $(t = 0.5)$	0.0	**0.4106**	1.0
u^2 $(t = 1.0)$	0.0	**0.4756**	1.0

Key Takeaway

The calculation for u_1^2 required **both** u_1^1 and u_1^0 to compute its history term H_1^1. This demonstrates the non-local memory of the fractional derivative.

Error Analysis: How Do We Know It's Correct?

The Challenge

To find the *true* error, we must compare our numerical solution u_j^n to the *exact* analytical solution $u_{exact}(x_j, t_n)$.

Problem: For this complex, nonlinear, fractional PDE, no simple exact solution is known!

Two Standard Methods

1. **Method 1: Error Norms (Ideal Case)** If we *did* have an exact solution, we'd measure the error with L_∞ or L_2 norms.
2. **Method 2: Rate of Convergence (Practical Case)** We can verify our code by proving it converges at the correct theoretical rate. This is the standard for research.

Error Analysis: Method 1 (Error Norms)

If we had an $u_{exact}(x, t)$, we would compute the error at each time step:

L_∞ Error (Maximum Error)

Tells you the **worst-case error** at any single point on the grid.

$$E_\infty^n = \max_j |u_{exact}(x_j, t_n) - u_j^n|$$

L_2 Error (RMS Error)

Gives a sense of the **average error** across the whole grid.

$$E_2^n = \sqrt{\Delta x \sum_{j=1}^{N-1} (u_{exact}(x_j, t_n) - u_j^n)^2}$$

Error Analysis: Method 2 (Rate of Convergence)

This is how we test our code without an exact solution.

The Theory

The error E is proportional to our step sizes: $E \approx C(\Delta t^p + \Delta x^q)$.

- The theoretical temporal order for our scheme is $\mathbf{p = 2 - \alpha}$.
- Since we used $\alpha = 0.5$, we must prove that $\mathbf{p \approx 1.5}$.

The Procedure

1. Run 3 simulations with a fixed, small Δx:
   - **Coarse:** $\Delta t_1 \rightarrow$ Get u_{coarse}
   - **Fine:** $\Delta t_2 = \Delta t_1/2 \rightarrow$ Get u_{fine}
   - **Finest:** $\Delta t_3 = \Delta t_1/4 \rightarrow$ Get u_{finest}
2. Treat u_{finest} as the "true" solution.
3. Calculate errors: $E_1 = |u_{finest} - u_{coarse}|$ and $E_2 = |u_{finest} - u_{fine}|$.

Convergence Rate: Simulated Example

We run the code (with a very fine Δx) and get the solution at $T = 1.0$.

Table: Simulated results for $u(0.5, 1.0)$ to find temporal order.

Run	Δt	Solution $u(0.5, 1.0)$	Error (vs. Finest)
Coarse	0.1	0.4615	$E_1 = \lvert 0.4688 - 0.4615 \rvert = 0.0073$
Fine	0.05	0.4662	$E_2 = \lvert 0.4688 - 0.4662 \rvert = 0.0026$
Finest	0.025	0.4688	— (Used as "truth")

Calculate the Order p

The error ratio should be $\frac{E_1}{E_2} \approx \left(\frac{\Delta t_1}{\Delta t_2}\right)^p = 2^p$.

$$p = \log_2\left(\frac{E_1}{E_2}\right) = \log_2\left(\frac{0.0073}{0.0026}\right) = \log_2(2.807) \approx \mathbf{1.49}$$

Conclusion

The numerical order $p \approx 1.49$ is extremely close to the theoretical order $p = 2 - \alpha = 1.5$. **This verifies our method is correct.**

Governing Equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad \text{(Heat Equation: } \alpha = 1, a = 0, c = 1)$$

Initial and Boundary Conditions:

- **Initial Condition:**

$$u(x, 0) = \sin(\pi x)$$

- **Boundary Conditions:**

$$u(0, t) = 0 \quad \text{and} \quad u(1, t) = 0$$

Exact Analytical Solution:

$$u_{exact}(x, t) = e^{-\pi^2 t} \sin(\pi x)$$

Numerical Approximation Results

t / x	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.0	0.0	0.309017	0.587785	0.809017	0.951057	1.000000	0.951057	0.809017	0.587785	0.309017	0.0
0.1	0.0	0.105928	0.201488	0.277324	0.326014	0.342791	0.326014	0.277324	0.201488	0.105928	0.0
0.2	0.0	0.036311	0.069068	0.095064	0.111755	0.117506	0.111755	0.095064	0.069068	0.036311	0.0
0.3	0.0	0.012447	0.023676	0.032587	0.038309	0.040280	0.038309	0.032587	0.023676	0.012447	0.0
0.4	0.0	0.004267	0.008116	0.011171	0.013132	0.013808	0.013132	0.011171	0.008116	0.004267	0.0
0.5	0.0	0.001463	0.002782	0.003829	0.004501	0.004733	0.004501	0.003829	0.002782	0.001463	0.0
0.6	0.0	0.000501	0.000954	0.001313	0.001543	0.001622	0.001543	0.001313	0.000954	0.000501	0.0
0.7	0.0	0.000172	0.000327	0.000450	0.000529	0.000556	0.000529	0.000450	0.000327	0.000172	0.0
0.8	0.0	0.000059	0.000112	0.000154	0.000181	0.000191	0.000181	0.000154	0.000112	0.000059	0.0
0.9	0.0	0.000020	0.000038	0.000053	0.000062	0.000065	0.000062	0.000053	0.000038	0.000020	0.0
1.0	0.0	0.000007	0.000013	0.000018	0.000021	0.000022	0.000021	0.000018	0.000013	0.000007	0.0

Absolute Error ($|u_{approx} - u_{exact}|$)

t / x	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.1	0.0000	0.0092	0.0176	0.0242	0.0285	**0.0299**	0.0285	0.0242	0.0176	0.0092	0.0000
0.2	0.0000	0.0066	0.0126	0.0173	0.0204	0.0214	0.0204	0.0173	0.0126	0.0066	0.0000
0.3	0.0000	0.0036	0.0068	0.0093	0.0109	0.0115	0.0109	0.0093	0.0068	0.0036	0.0000
0.4	0.0000	0.0017	0.0032	0.0044	0.0052	0.0055	0.0052	0.0044	0.0032	0.0017	0.0000
0.5	0.0000	0.0008	0.0014	0.0020	0.0023	0.0025	0.0023	0.0020	0.0014	0.0008	0.0000
0.6	0.0000	0.0003	0.0006	0.0009	0.0010	0.0011	0.0010	0.0009	0.0006	0.0003	0.0000
0.7	0.0000	0.0001	0.0003	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003	0.0001	0.0000
0.8	0.0000	0.0001	0.0001	0.0001	0.0002	0.0002	0.0002	0.0001	0.0001	0.0001	0.0000
0.9	0.0000	0.0000	0.0000	0.0001	0.0001	0.0001	0.0001	0.0001	0.0000	0.0000	0.0000
1.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Max Error: **0.0299** at $t = 0.1, x = 0.5$

Visual Validation: Exact vs. Numerical Solution

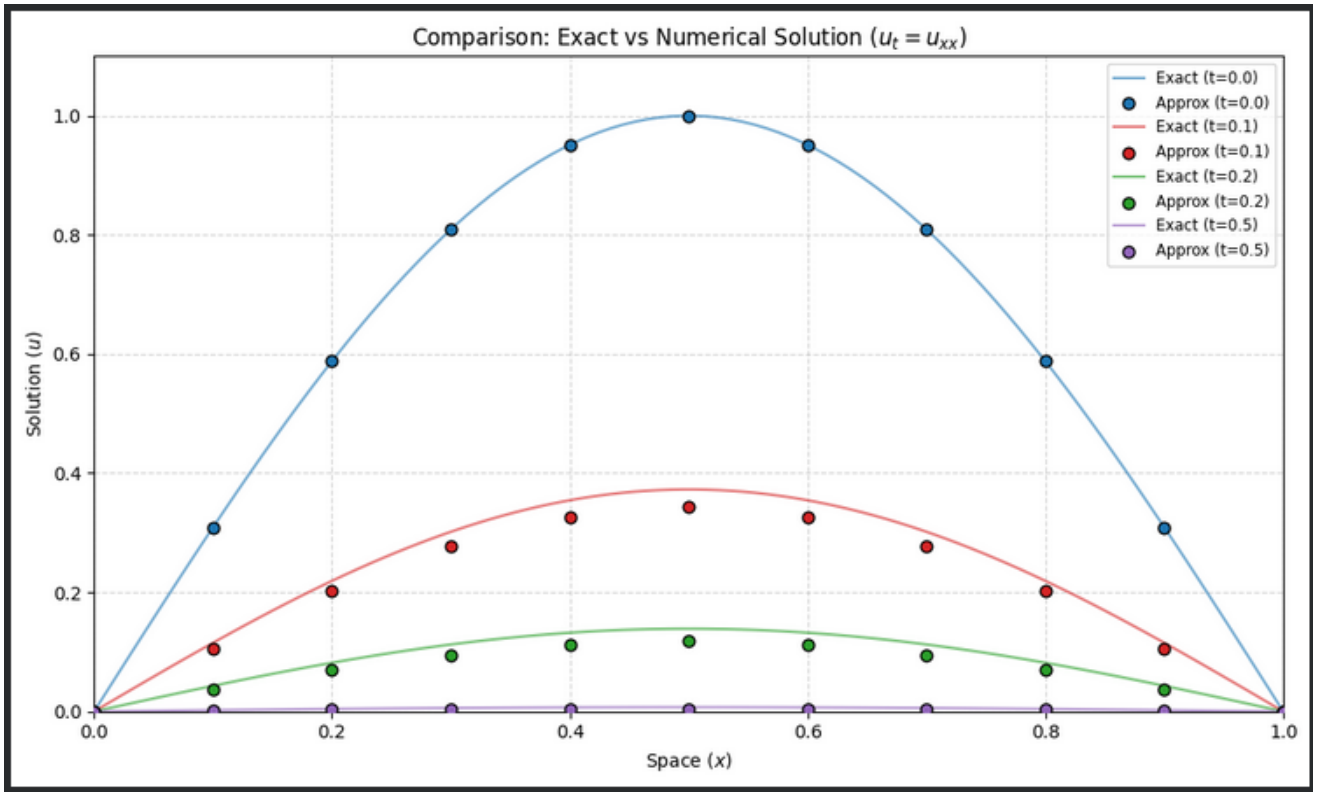

Figure: Spatial distribution of $u(x, t)$ at selected time steps.

- **Initial State ($t = 0.0$):** The numerical approximation (blue dots) aligns perfectly with the exact sine wave, verifying the initial condition.
- **Transient Error ($t = 0.1$):** A visible deviation occurs at the peak (red dots vs. line), corresponding to the max error (≈ 0.0299) identified in the error matrix.
- **Physical Behavior:** Despite the discretization error, the numerical solution correctly captures the *diffusive decay* behavior as time progresses ($t = 0.2, 0.5$).

Visualizing the Transition to Integer Order

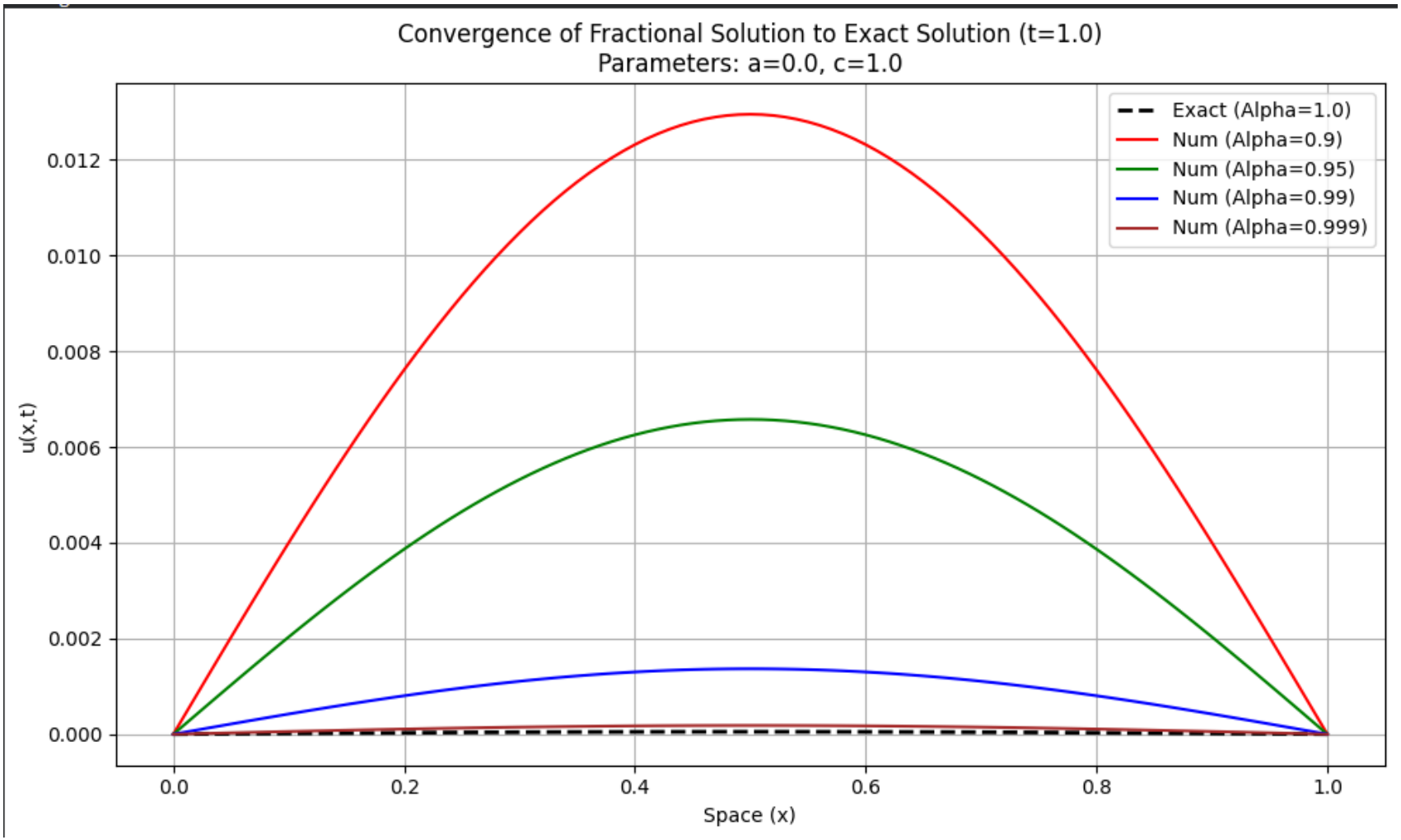

Figure: Numerical solutions for $\alpha \in \{0.9, 0.95, 0.99\}$ vs. Exact Solution ($\alpha = 1$) at $t = 1.0$.

Convergence Analysis: Discussion

The Limit $\alpha \to 1$

We investigated the behavior of the numerical solution as the fractional order α approaches the classical limit of 1.

- **Setup:** We solved the case $(a = 0, c = 1)$ where the exact solution for $\alpha = 1$ is known (Heat Equation).
- **Observation:** As seen in the previous slide, as α increases from $0.9 \to 0.99$, the numerical curves (colored) progressively rise and tighten to match the exact classical solution (black dashed line).

Key Insight

As $\alpha \to 1$, the numerical solution converges to the classical integer-order solution. This confirms the mathematical consistency of the fractional model with the standard limit.

Conclusion

- The L1/Crank-Nicolson scheme is a robust and stable method for solving the nonlinear fractional Burgers' equation.
- The **L1 formula** correctly handles the non-local "memory" of the fractional derivative.
- The **Predictor-Corrector** step efficiently linearizes the auu_x term.
- The final problem reduces to solving a **tridiagonal system** at each time step, which is computationally fast.
- **Convergence rate analysis** is the standard method to verify the correctness of the implementation when an exact solution is not known.

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